MILITARY
SPECIFICATION

MIL-C-55120(SigC)
14 March 1961

CAMERA, STILL PICTURE, KA-30( )

1. SCOPE

1.1 Scope.- This specification covers one type of aerial reconnaissance camera, designated as Camera, Still Picture, KA-30( ). (See 6.5).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire and Solid Form.

MILITARY

MIL-F-32	Film, Photographic, Aerial, Black and White
MIL-P-116	Preservation, Methods of.
MIL-V-173	Varnish, Moisture and Fungus Resistant for Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Non Rigid.
JAN-F-675	Films, Reflection-Reducing, for Glass Optical Elements.
MIL-T-713	Twine and Tape, Lacing and Tying (For use in Electrical and Electronic Equipment).
MIL-I-3190	Insulation Sleeving, Electrical, Flexible, Treated.
MIL-C-4150	Case, Carrying and Storage, Shock and Waterproof.
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-I-6181	Interference Control Requirements, Aircraft Equipment.

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MIL-W-6858	Welding: Aluminum, Magnesium, Nonhardening Steels or Alloys, and Titanium, Spot, Seam and Stitch.
MIL-I-7444	Insulation Sleeving, Electrical, Flexible.
MIL-I-7798	Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic.
MIL-B-7883	Brazing of Steels, Coppers, Copper Alloys, and Nickel Alloys.
MIL-E-7894	Electric Power, Aircraft, Characteristics of.
MIL-T-7928	Terminals; Lug Crimp Style, Copper, Insulated.
MIL-F-9329	Filters, Light, Photographic Lens, for Aerial and Ground Cameras, General Specification for.
MIL-M-10578	Metal Conditioner and Rust-Remover (Phosphoric and Base).
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).
MIL-T-15659	Terminals; Lug Solder Type, Copper.
MIL-I-18057	Insulation Sleeving, Electrical, Flexible, Glass Fiber, Silicone Rubber Treated.

STANDARDS

MILITARY

MIL-STD-22	Welded Joint Designs.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-150	Photographic Lenses.
MIL-STD-152	General Environmental Requirements, Testing Procedures, and Sampling for Aerial Photographic Reconnaissance Equipment.
MIL-STD-180	Photographic Equipment, Classification of Visual and Mechanical Defects for.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-681	Identification Coding and Application of Hookup Wire.
MS-3106	Connectors-Electric Plugs, Straight.
MS-33523	Film; Standard Dimensions for Nominal 5-inch Width, Unperforated, Aerial.
MS-35436	Terminal, Lug, Solder Type, Copper Stamping, Insulation Grip, One Hole.
MS-35440	Terminal, Lug, Solder Type, Copper Stamping, Insulation Grip, Slotted.

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AIR FORCE-NAVY AERONAUTICAL

AN 7023 Connector, Vacuum Hose, Photographic

DRAWINGS

SIGNAL CORPS

SC-A-46420	Fasteners.
SC-A-46469	Lubricants Guide.
SC-A-362100	Preparation of Preproduction Packs.

U. S. AIR FORCE

51C17848	Spool, Aerial Roll Film 5-inch (100 ft.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description.- Camera, Still Picture, KA-30( ) shall produce high quality day and night negatives, a nominal 4.5 by 4.5 inch, on 5 inch wide aerial roll film, when mounted in a high speed reconnaissance aircraft. The camera shall be capable of operation with a Mohawk Photo System Unit, hereafter referred to as the camera control system with all necessary switching functions, diaphragm and shutter speed selections being performed by the control system. Image motion compensation (IMC) shall be provided for both day and night photography. The camera shall be suitable for vertical and side oblique mounting and shall accept various focal length lens cones. When positioned in any oblique mounting, the camera shall be capable of providing 80 percent overlap photography with the correct IMC rate on the optical axis of obliquity as specified herein.

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3.1.1 Major components.- The major components of Camera, Still Picture, KA-30( ), shall be as listed below.

Component	Quantity Required	Paragraph
Camera body LA-134( )	1	3.5
Cassette, film (100 ft capacity LA-135( )	2	3.8
Lens Cone, 6-inch focal length LA-131( )	1	3.6

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples of Camera, Still Picture, KA-30( ) for approval provided the invitation for bids and contract requires samples. (See 4.3.)

3.3 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.4 Carrying device.- A suitable device shall be provided for hand carrying the assembled camera. The carrying device, such as a handle or strap, shall be easily but securely attached to the assembled camera by a hook, thumb screw or similar means.

3.5 Camera body, LA-134( ).- The camera body shall be the principal member of the camera. The camera body shall contain the drive mechanism for film transport and film control and shall provide for proper attachment of the lens cone and film cassettes. The camera body shall contain a suitable mechanism for maintaining film flatness during exposure when connected to the regulated vacuum supply in the aircraft. A permanently legible arrow with the word "FLIGHT" shall be placed on the side of the camera body to indicate the required direction of flight.

3.5.1 Drive mechanism.- The drive mechanism shall be driven by a variable speed motor, which shall operate the film transport mechanism and provide the modes of operation required of the camera. The drive mechanism shall have four electric clutches which shall be driven at different speed ratios with reference to the motor, to produce the various modes of operation. The clutches shall be used to drive the film metering roller as required to control the linear velocity of the film.

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3.5.2 Film transport mechanism.- The film transport mechanism shall provide the means for automatically advancing, properly spacing, accurately tracking, and holding the film. The film transport mechanism shall include reliable provision for threading the film through the camera body without separation of the camera body from the lens cone. The mechanism shall give regular and positive film movement throughout the life of the camera.

3.5.3 Focal plane plate.- The camera body shall have an accurately machined focal plane plate (platen) as specified in Figure 1, for minimizing irregularities of the film in the format area, during exposure. The surface of the focal plane plate shall be precisely aligned with the lens cone mating surface to satisfy the camera system resolution specified herein when the lens cone specified herein is attached to the camera body.

3.5.4 Format size.- The camera body shall be provided with a format mask, located at the focal plane, which shall have a picture aperture of 4.5 by 4.5 inches, plus or minus 0.005 inch, a data aperture and four fiducial markers as specified in Figure 2.

3.5.5 Film vacuum system.- The camera body shall contain the vacuum regulator and controls necessary to hold the film flat during exposure. The camera body regulator shall be supplied from an external vacuum source. The vacuum source shall supply vacuum from 1 to 8 inches of mercury and a minimum flow of 0.5 cubic feet per minute. An AN7023 fitting shall be provided on the outside of the camera body for connection to the vacuum source.

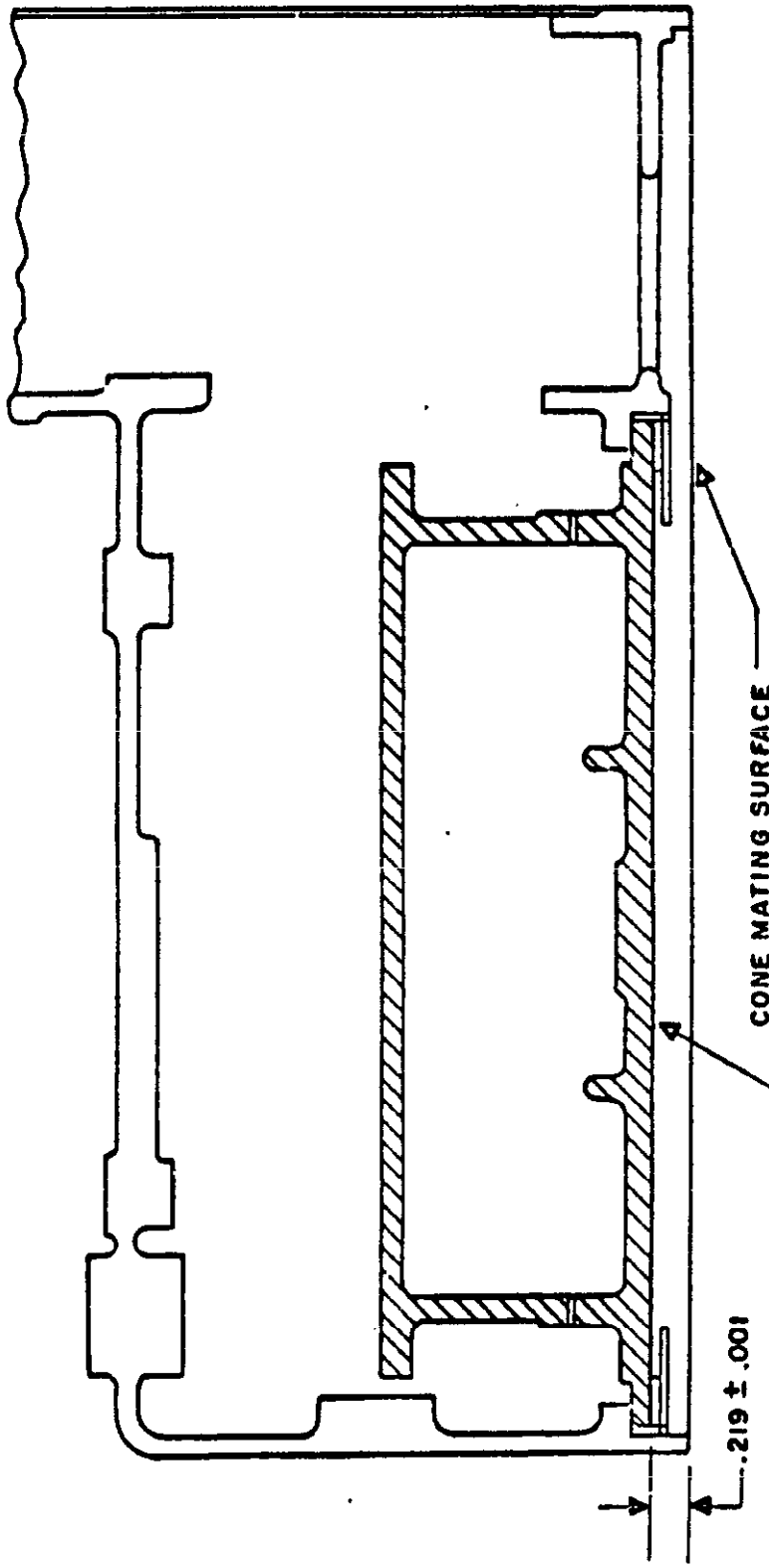
3.5.6 Cycle counter.- The camera body shall have a five digit, additive non-resettable counter for automatically counting the cycles undergone by the camera. A cycle consists of one complete operation of the shutter assembly.

3.5.7 Mode selector.- The camera body shall be provided with a mode selector that is operable with a screwdriver or similar device. The mode selector shall perform the necessary mechanical and electrical switching on the camera as required to perform the selection of mode of operation as specified herein.

3.5.7.1 Remote mode selection.- One position on the camera mode selector, in conjunction with the applicable control system and a cockpit selector switch, shall provide for remote or cockpit selection of camera operating modes as defined herein.

3.6 Lens cone-6-inch focal length, LA-131( ).- The lens cone shall rigidly attach to the camera body but shall be removable when desired. Attachment of the lens cone to the camera body shall automatically engage all mechanical and electrical connections necessary for proper functioning of mechanism contained in the cone. The lens cone shall include the lens, shutter, exposure control servo, and data recording unit. The lens shall extend sufficiently to protect the lens and filter.

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All dimensions are in inches

FIGURE 1 PLATEN TOLERANCES, CAMERA BODY, LA-134( )

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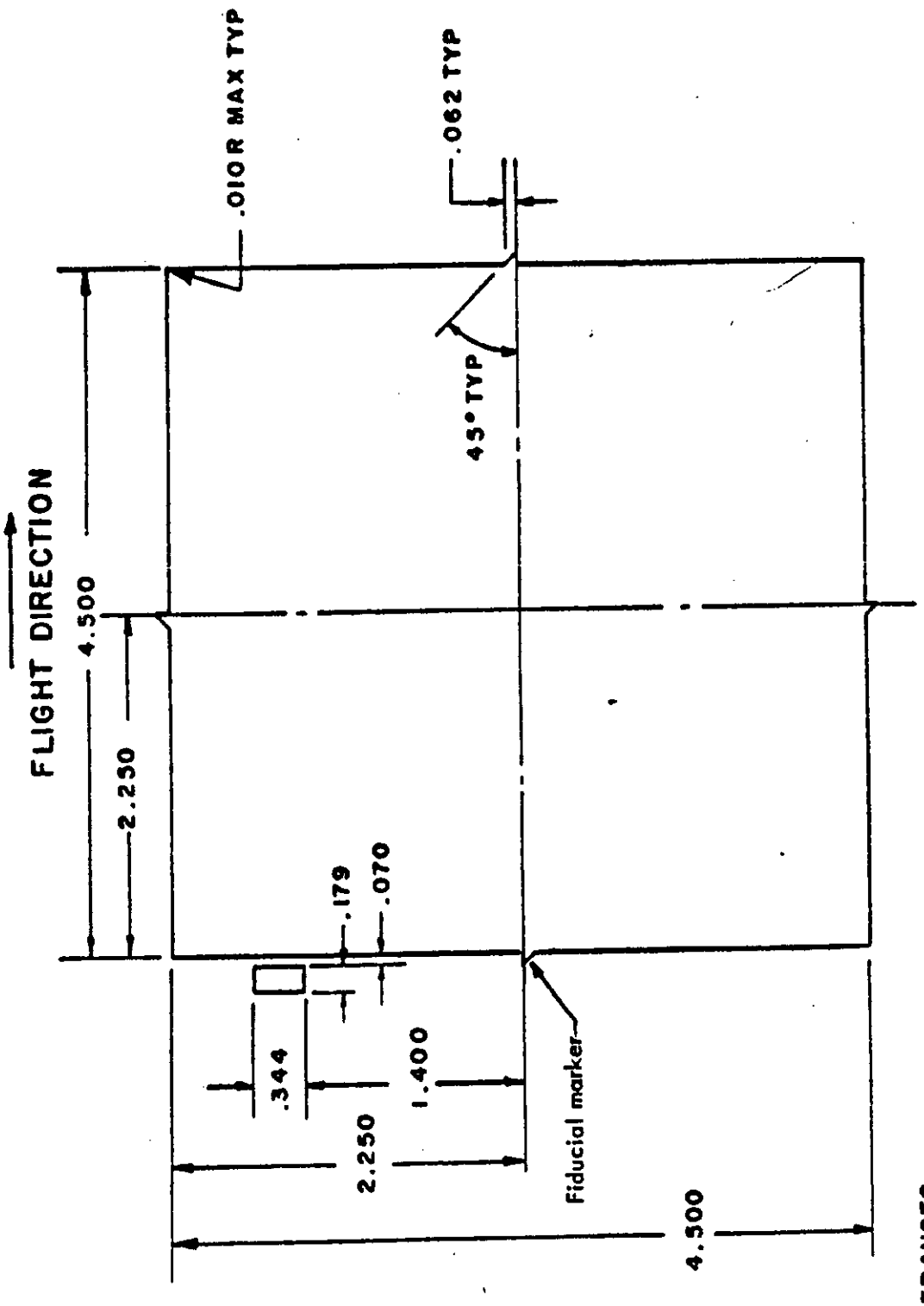


FIGURE 12 FORMAT MASK, FIDUCIAL MARKERS, AND DATA LOCATION, CAMERA BODY, LA-134( )

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3.6.1 Shutter.- The shutter shall be of the uni-directional between- the- lens type. The shutter shall be suitable for both day and night modes of operation as specified herein. The shutter shall be designed to minimize any possible degradation of system resolution.

3.6.1.1 Shutter speeds.- The shutter shall have continuously variable effective speeds from 1/50 second to 1/400 second. The effective shutter speed shall be within plus or minus 25 percent of that which is marked on the shutter speed dial. At any shutter speed setting, the effective shutter speed shall not vary more than plus or minur 15 percent during a consecutive series of exposures. (See 4.21).

3.6.1.2 Shutter release.- The shutter shall be provided with the means to permit it to be manually tripped to facilitate opening the shutter and to prepare the lens cone for storage.

3.6.1.3 Diaphragm.- The shutter assembly shall include an iris diaphragm which shall be calibrated for the appropriate lens that it will mount. The diaphragm scale shall have conventional f/number calibrations in accordance with MIL-STD-150.

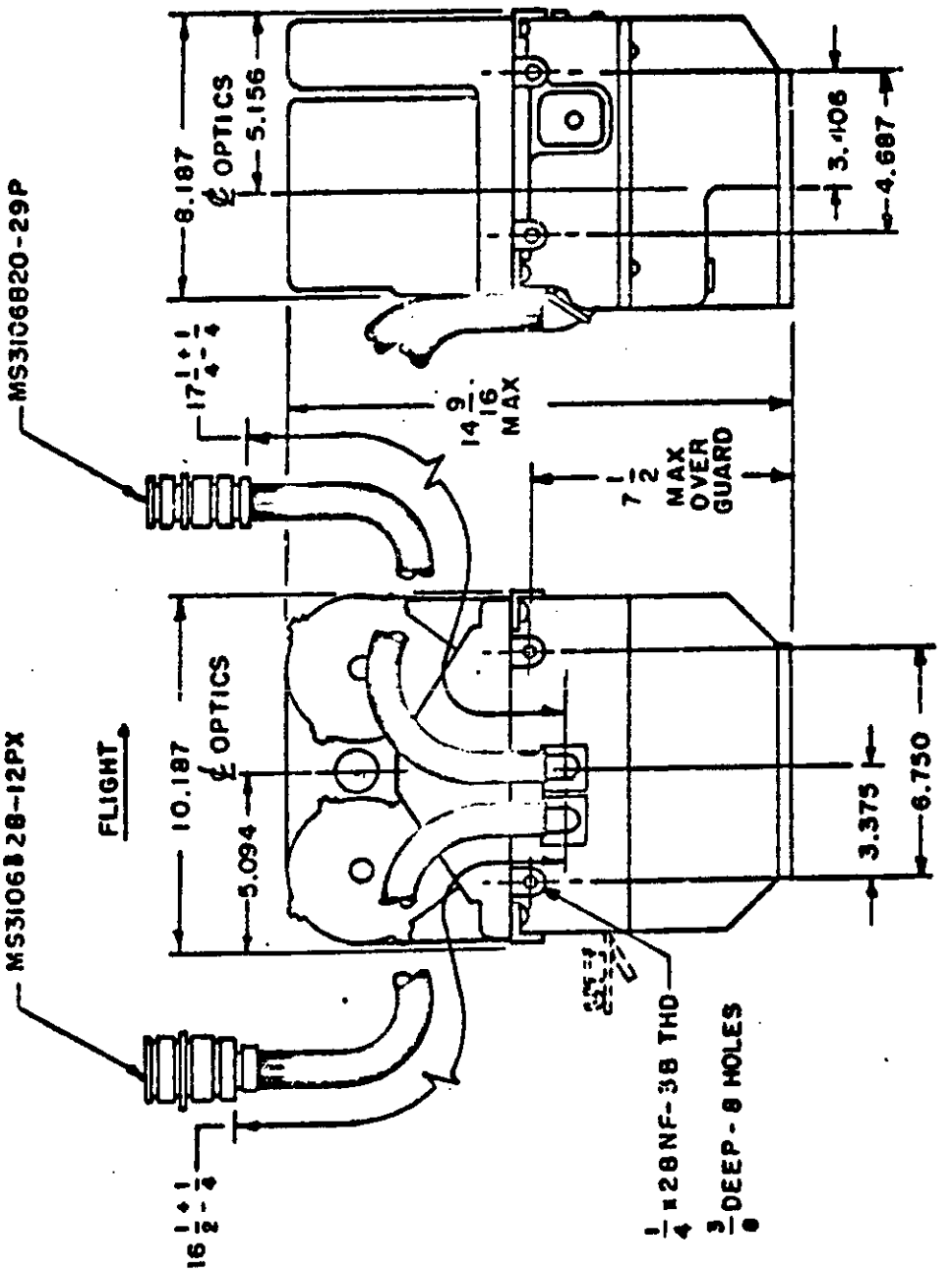
3.6.1.4 Cycle counter.- The shutter assembly shall have a five digit, additive, nonresettable counter for automatically counting the number of cycles of operation the shutter has performed.

3.6.2 Recording unit.- The lens cone shall contain a recording unit which includes a 24 hour clock with sweep second hand and a data card. Provision shall be made for removing the recording unit for operation of the camera without the recording unit. The recording unit shall be positioned to record the data between exposure frames. An arrow on the data card shall point to the appropriate frame. All data on the data card, which includes the arrow and any written data, and also the numerals and position of the hands of the clock, shall be readable from the negative without any visual aids after processing of the film.

3.6.3 Mounting provision.- The lens cone shall provide suitable pads for mounting the camera in an aircraft. The dimensions of the mounting pads shall be as specified in Figure 3. .

3.6.4 Pendant cables.- The lens cone shall have two pendant cables for connecting the camera to the control system. Connectors and their pin functions shall be as follows:

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All dimensions are in inches

FIGURE 3 - MOUNTING DIMENSIONS, AERIAL CAMERA KA-30( ), WITH 6-INCH LENS CONE

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3.6.4 Pendant cables .-(Cont'd)

Primary Connector (MS3106B-28-12PX)

A - Power Ground
B - + 28 VDC Power
C - (Spare)
D - Signal Ground
E - Exposure reference
F - Exposure feedback
G - Interlock + 28 volts
H - (+) Tachometer
J - (-) Tachometer
K - AC Power (II)
L - (+) Film Drive
M - (-) Film Drive
N - Exposure Drive
P - (Spare)
R - (Spare)
S - Shutter close
T - (Spare)
U - (Spare)
V - (Spare)
W - Shield Ground
X - (Spare)
Y - (Spare)
Z - (Spare)
a - Operate
b - Cycle
d - Operate and Fail Indicate

Secondary Connector (MS3106B-20-29P)

A - Power Ground
B - Day Mode Switch
C - Sync
D - IMC Pulse Mode
E - Night Mode
F - Night Interlock + 28 VDC
G - (Spare)
H - (Spare)
K - 5.272 V Tachometer
J - Extra picture Interlock II
L - DC Exposure Decrease
M - DC Exposure Increase
N - Extra Picture Interlock I
P - Extra Picture
R - Extra Picture Drive
S - (Spare)
T - (Spare)

3.6.5 Exposure control servo.- The lens cone shall include an exposure servo positioner for adjusting the lens diaphragm and the shutter time to produce an effective exposure corresponding to the control system input as specified in paragraph 3.26.1.2. Provision shall be made for manually operating the servo positioner.

3.6.5.1 Exposure Feedback accuracy.- At any position of the exposure indicator, the exposure feedback voltage shall be within plus or minus 1.5 volts of the nominal feedback voltage to be determined as specified in paragraph 3.26.1.2. (See 4.13).

3.6.5.2 Manual S/C inputs.- The exposure control servo shall include a provision for manually setting S/C during preflight where:

S - Film Sensitivity in exposure index values
C - Filter-Factor correction

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3.6.6 Exposure indicator.- The lens cone shall have an externally visible exposure indicator. The indicator shall be calibrated to indicate the effective shutter speed and diaphragm setting.

3.6.7 Lens, 6-inch focal length.- The lens shall be suitable for use in aerial reconnaissance and shall be Type I, in accordance with Standard MIL-STD-150. All air to glass surfaces of the lens shall be anti-reflection coated in accordance with JAN-F-675. The relative aperture range of the 6-inch focal length lens shall be f/2.8 to f/16.

3.6.7.1 Lens resolution.- The 6-inch lens shall have an Area Weighted Average Resolution (AWAR) of no less than 26 lines/mm, with a minimum of 14 lines/mm at any position within the format area, when tested in accordance with Method 11 of Standard MIL-STD-150. (See 4.9.)

3.6.7.2 Lens cap.- The lens assembly shall be provided with a front and rear lens cap. The front lens cap shall securely attach to the bayonet filter mount provision. The rear lens cap shall be a snug fitting slip or snap on type.

3.6.7.3 Lens focusing.- Each lens, when installed in a lens cone, shall be checked over a range of focal plane locations as required to determine the focal plane location of best AWAR within .002 inches for one diagonal across the format. After the focal plane location of best AWAR has been determined for one diagonal the AWAR of the opposite diagonal shall be checked in the same focal plane. The AWAR of each of the two diagonals must be at least 89 percent of the AWAR shown on the lens manufacturer's Lens Performance Data Sheet. In the event that the second diagonal does not produce the required AWAR, a new focal plane location shall be found which will yield the required AWAR on both diagonals. A record of the resulting AWAR and the focal plane location of best AWAR for each lens cone shall be retained in the manufacturer's inspection records.

3.7 Filters.- Two filters shall be furnished with each lens. One shall be equal to a Wratten No. 25 (Red), the other shall be equal to a Wratten 12 (Yellow) in accordance with Class III of Specification MIL-F-9329. Each filter shall be mounted in a bayonet, lock-type filter mount and shall properly and firmly attach to the lens.

3.8 Film cassettes.- The camera shall be supplied with two film cassettes, one for feed and the other for take-up of film. The film cassettes shall be mutually interchangeable. That is, both shall be equally suitable for either film feed or film take-up. The cassettes shall rigidly attach to the camera body but shall be readily removable for loading or unloading. Attachment of the cassettes shall automatically engage all mechanical and electrical connections necessary for proper operation of the mechanism contained in the cassettes.

3.8.1 Film footage indicator.- Each cassette shall be equipped with a film footage indicator. The indicator shall indicate the amount of film remaining in the cassette.

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3.8.2 Magazine, film, cassette type LA-135( ).- The film cassettes supplied with the camera shall accomodate 100 foot lengths of 5-inch wide aerial roll film as specified in MIL-F-32 conforming to Standard MS33523 and wound on spools conforming to Drawing 51C17848-2 (USAF). The cassette shall be designated as Magazine, Film, Cassette Type LA-135( ).

3.8.3 Magazine, film, cassette type LA-141( ).- Camera, Still Picture, KA-30( ), shall be capable of operating, without modification, with 250 foot cassettes attached. The cassette is designated as Magazine, Film, Cassette Type, LA-141( ), and shall accomodate the 5-inch film specified in 3.8.2 wound on spools conforming to Drawing 51C17848-3 (USAF).

3.9 Carrying cases LA-138( ) and LA-139( ).- Two carrying cases, in accordance with Specification MIL-C-4150, shall be supplied. Carrying Case LA-138( ) shall house Lens Cone LA-131( ) and accessories and Carrying Case LA-139( ) shall house Camera Body LA-134( ), Cassettes, LA-135( ) and accessories.

3.10 Screws, other threaded devices, and related parts.-

3.10.1 Form and fit.- Unless otherwise specified, screw threads shall conform to Handbook H28; the Unified Thread Form and Series shall be class 2A or 2B fit, and the American National Thread Series shall be class 2 fit.

3.10.2 Machine screws, cap screws, and washers.- When practicable, machine screws, cap screws, and washers shall be selected from Drawing SC-A-46420.

3.10.3 Self-tapping screws.- Thread-forming screws shall not be used. Thread-cutting screws shall be used only if construction is improved thereby and only in places where the screws will not require loosening or removal during operation of maintenance. In addition, their use shall be subject to approval by the contracting officer. Chips formed in through holes by thread-cutting screws shall be removed.

3.11 Cleaning.-

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.11.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.12 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting.

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3.13 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4).

3.13.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.13.2 Serial numbers.- The following items shall have serial numbers:

Body, Camera, LA-134( )
Lens Cone Assembly, 6 inch focal length LA-131( )
Cassette (100 ft. capacity) LA-135( ).

3.14 Riveting.- Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivets holes to the edges of the material in which the holes are formed shall be at least 1-1/2 times the rivet diameter. Rivets shall be used where practicable in preference to other means of securing aluminum or magnesium structural parts if the joint is subjected to dynamic tension loads when the equipment is exposed to specified service conditions and if the parts do not require removal when the equipment is serviced. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified shock test. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.16 Soldering and brazing.-

3.16.1 Brazing.- Brazing of steels, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.16.2 Soldering.-

3.16.2.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

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3.16.2.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.16.2.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall have become loosened.

3.17 Tropicalization of material.- Material which is a nutrient for fungus growth shall be treated as follows:

3.17.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.17.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.17.3 Flexibility.- Treatment shall not effect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.17.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2.)

3.18 Welding.- Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

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3.18.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.18.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas- shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.18.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.19 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.19.1 Cabling.- Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713 or tape conforming to Specification MIL-I-7798. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.19.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected.

(d) To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement.

(e) To facilitate field repair of broken or cut wires.

(f) To prevent chafing or breaking of individual wires due to flexing of hinged parts.

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3.19.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces to sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields, or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 1/8 inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.19.4 Support.- Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner. (NOTE: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.) Twine or tape shall not be used for securing wire and cable.

3.19.5 Clearance.- Clearance between wires or cables, and parts such as resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions. For electrical clearance and creepage distances see 3.19.11.

3.19.6 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.19.7 Insulating sleeving.- Insulating sleeving shall not be used unless specified for a particular application or specifically approved by the contracting officer. Approval will not be granted when other means of insulation are practicable. If insulating sleeving is used, it shall conform to Specification MIL-I-631, MIL-I-3190, MIL-I-7444, or MIL-I-18057, as applicable.

3.19.8 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

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3.19.8.1 Terminal lugs.- Terminal lugs shall be of the type that grips the wire insulation. Solderless terminal lugs shall conform to Specification MIL-T-7928. Solder-type terminal lugs shall conform to Specification MIL-T-15659 and shall be of the types shown on Standard MS 35436 and MS35440.

3.19.8.2 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928, except that the test requirements shall conform to the table I.

TABLE I

Test requirements for solderless-terminal-lug connections

Wire size AWG	Test current (amperes)	Max voltage drop (millivolts)		Min strength (pounds)
		Initial	After test	
22	9	8	13	15
20	11	7	12	19
18	16	7	12	36
16	22	7	12	50
14	32	6	11	70
12	41	5	8	110
10	55	5	8	180

3.19.9 Grounding.- Ground connection to shields and to other mechanical parts, except the frame, shall not be made to complete electrical circuits but only to eliminate high-potential a.c. points.

3.19.10 Identification of wiring.- Wiring shall be identified by color-coding or, if approved by the contracting officer, may be identified by other means such as marking of terminals at both ends of each lead. Color-coding for wiring shall conform to Standard MIL-STD-681.

3.19.11 Electrical clearance and creepage distances.- Minimum electrical clearances and creepage distances shall conform to table II.

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TABLE II

Electrical clearance and creepage distances (in inches)
(Except for printed circuits)

Voltage, ac or dc	Clearance	Creepage
To 300	1/16	1/16
300 -600	1/16	1/3

3.20 Dimensions.- The dimensions of the camera body, film cassettes, and the 6-inch lens cone shall be as specified in Figure 4.

3.21 Weight.- The total weight of the camera body, two empty 100 foot film cassettes, and the 6-inch lens cone shall not exceed 35 pounds.

3.22 Electrical requirements.- Camera, Still Picture KA-30( ) shall operate from a combination of AC and DC power as specified herein.

3.22.1 Input power requirements.- The camera shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The current drain shall not exceed the specified amounts.

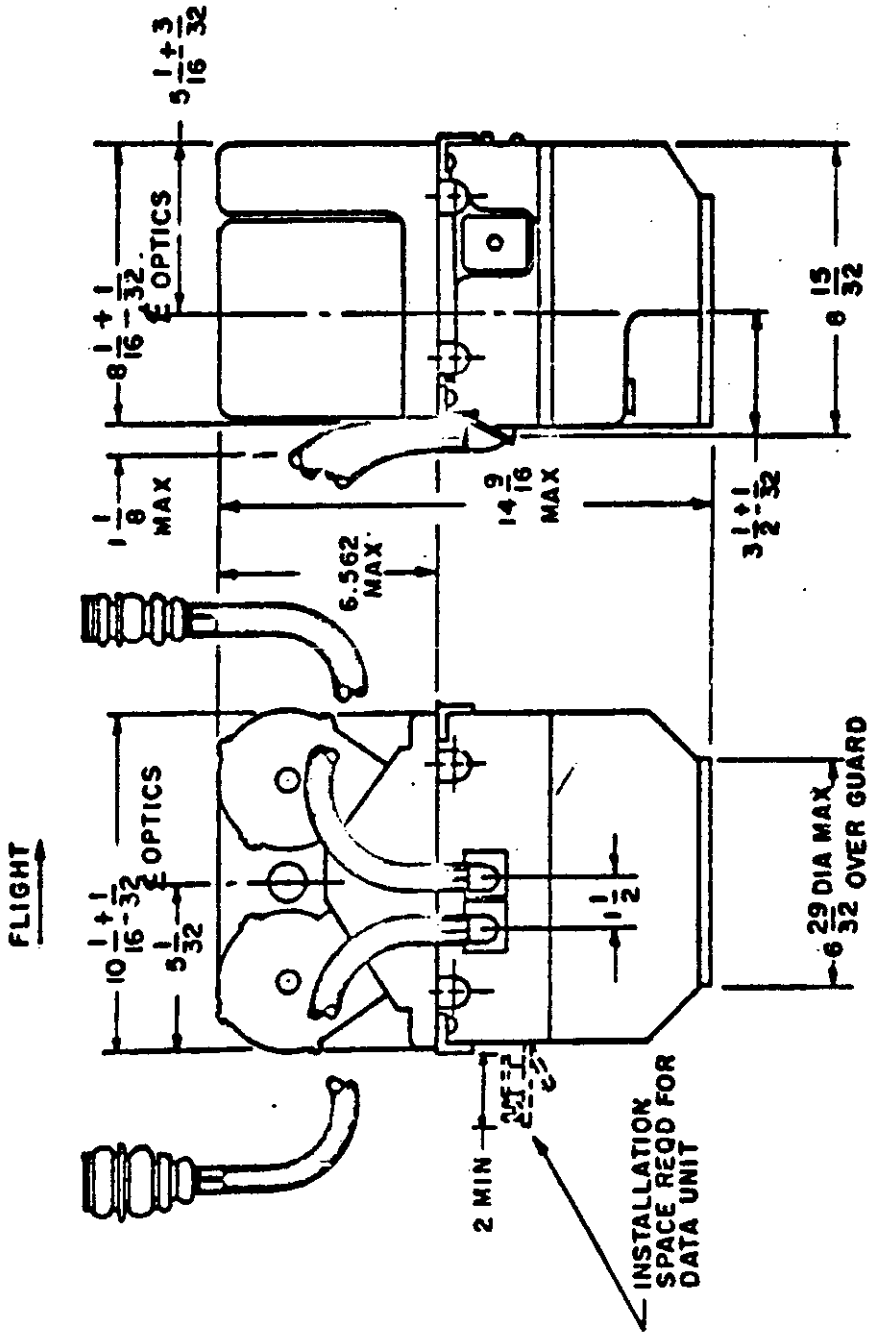
(1) AC Power (Single Phase) 115 Volt

Operating Voltage Limits	102 to 124 volts
Operating Frequency Limits	380 to 420 cps
Current drain	0.200 amp (For 100 foot cassette)

(2) DC Power 28 Volt

Operating Voltage Limits	25 to 29 Volts
Current drain	5.2 Amps (For 100 foot cassette)

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ALL DIMENSIONS ARE IN INCHES

FIGURE 4 - MAXIMUM DIMENSION, AERIAL CAMERA KA-30( ), 100-FEET FILM CAPACITY, 6-INCH f/2.8 LENS

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3.22.1.1 AC Frequency.- The equipment shall be designed for operation over the frequency range of 380 - 420 cps. Operation down to 360 cps for limited periods of time shall result in no damage to equipment. The equipment shall resume the normal operation when the frequency returns to within the 380 - 420 cps limits.

3.22.2 Film safety switch.- A switch shall be provided within the camera to shut off the electrical power to the film drive motor and shutter trip circuit at the time the film supply is exhausted or when the film cassettes are improperly seated. The switch shall also shut off the 28 volt DC power from the camera interlock lead, and provide continuous indication of camera failure.

3.23 Service conditions.- The equipment shall operate satisfactorily and withstand storage under the specified conditions to meet the Environmental Requirements and Testing Procedures of MIL-STD-152 as stated, with the exception given herein.

3.23.1 Equipment Operating

a. Temperature	-As specified (0 to 120F)
b. Altitude	-As specified (up to 50,000 ft.)
c. Humidity	-As specified (5 to 95% except relative humidity shall be 30% for temperatures from 95 to 120 F and the camera shall not be required to take pictures under conditions wherein condensation takes place in or on the equipment.
d. Vibration	-As specified
e. Acceleration	-As specified (4G)
f. Radio Interference	Range 0.15 to 150 mc in lieu of 0.15 to 400 mc in MIL-I-6181
g. Explosion Proof	-As specified

3.23.2 Equipment Nonoperating

a. Temperature	-As specified (minus 65 to plus 160 F)
b. Altitude	-As specified (up to 50,000 ft.)
c. Humidity	-As specified (5% to saturation)
d. Fungus	-As specified
e. Sand and Dust	-Exposure to sand and dust will not be required
f. Shock	-As specified (7.5G)
g. Acceleration	-As specified

3.23.3 Attitude.- The equipment shall operate in any attitude or position.

3.23.4 Drip proof.- The camera shall be constructed so that when completely assembled for operation, water falling on the equipment shall not enter the assembly. (See 4.10).

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3.24 Modes of operation.- The camera, when supplied with electrical pulses from the camera control system or the test control unit specified herein (see 3.25), shall be capable of operating in each of four preselected modes (1) autocycle, (2) pulsed without IMC, (3) pulsed with IMC, and (4) open shutter (night). Film flatness shall be maintained during exposure by vacuum.

3.24.1 Autocycle mode.- When operating with full vacuum and a loaded cassette, the camera shall provide a maximum of 6 exposures per second with an image motion compensating (IMC) rate variable from 0.15 to 10.8 inches per second sychronized for 60 percent overlap in vertical photography, using either the 100 foot or 250 foot film cassette.

3.24.2 Pulsed without IMC mode.- In the non-IMC pulsed mode, the camera shall function as a simple pulse camera, producing one exposure and recycle for each actuating pulse received from the camera control system. The film shall move only during the recycle period. When operating with full vacuum and a loaded cassette in this mode, the camera shall provide a maximum of two exposures per second with a film capacity up to 100 feet and one exposure per second with a film capacity up to 250 feet.

3.24.3 Pulsed with IMC mode.-

3.24.3.1 100 foot film cassette.- The camera shall provide a maximum of three exposures per second with an image motion compensating rate variable from 0.15 to 10.8 inches per second with film capacity up to 100 feet. In this mode of operation, the camera shall be capable of providing up to 60 percent overlap in vertical photography with the correct IMC up to 5.4 inches per second on the optical axis. The camera shall be capable of providing up to 80 percent overlap when positioned in any oblique mounting with the correct IMC up to 2.7 inches per second on the optical axis. The exposure rate capability shall be obtained with the use of a loaded cassette and full vacuum.

3.24.3.2 250 foot film cassette.- The camera shall provide a maximum of two exposures per second with an image motion compensating rate variable from 0.15 to 3.6 inches per second with film capacity up to 250 feet. In this mode of operation, the camera shall be capable of providing up to 60 percent overlap in vertical photography with correct IMC up to 3.6 inches per second on the optical axis. The camera shall be capable of providing up to 80 percent overlap when positioned in any oblique mounting with the correct IMC up to 1.8 inches per second on the optical axis.

3.24.4 Open shutter (Night).- The camera, when used for night photography shall operate normally with the shutter open. The shutter shall close after illuminant burst, then recycle to open position. The cycle rate, IMC range and the overlap capabilities shall be identical to the IMC pulsed mode as specified in paragraph 3.24.3.

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3.24.5 Emergency operation.- The camera, when supplied with a 28 volt DC overriding command control system, shall produce exposures at a rate of not less than one exposure per second for the duration of the overriding control, while operating in any of the specified modes, except open shutter (night) mode.

3.24.5.1 Override open shutter (Night).- The camera shall produce a normal open shutter (night) cycle in conjunction with the control system provisions for overrun during night operation.

3.25 Performance.- The camera shall operate in three day modes and one night mode and shall be capable of being controlled by a Mohawk Photo System Unit, (JANCCS camera control system) , which may be Government furnished. If a control system is not Government furnished, the contractor shall be required to furnish a test control unit, which will simulate the operations of the Mohawk Photo System Unit. The lens diaphragm setting and the shutter setting shall be controlled from the aircraft camera control system. Operation of the remote exposure control circuit shall be evident by external visual observation of the camera during preflight check of the camera and camera control system. The preflight check will be made with the exposure control set for Speed Group 80 Film, and a Minus Blue filter factor. The resulting pictures made under the test conditions specified herein shall be of a quality acceptable to the requiring activity.

3.25.1 Cycling rate.- The camera shall have a cycling rate from 1 frame in 12 seconds to 6 frames per second in autocycle mode only.

3.25.2 Image motion compensation rate.- The image motion compensation rate shall be variable from 0.15 to 10.8 inches per second; for any camera tachometer output voltage, the IMC rate produced shall be equal to the nominal rate specified herein within a tolerance uniformly varying from plus to minus 5 percent at 10.8 inches per second IMC to plus or minus 10 percent at 0.15 inches per second IMC.

3.25.3 Frame spacing.- The distance between centers of any two consecutive exposure frames, including the space between frames, shall be as specified herein, except that when the camera changes modes of operation or is turned on at a high rate, the tolerances do not apply for the first three frames. The film exposure frames must not overlap at any time during any mode of operation.

Modes of Operation	Frame Spacing. Inches
a. Autocycle	4-7/8 plus or minus 1/16
b. Pulsed without IMC	
100 foot cassette	4-7/8 minus 1/16 to plus 5/8
250 foot cassette	4-25/32 minimum
c. Pulsed with IMC	4-13/16 minimum
d. Open shutter (night)	4-13/16 minimum

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3.25.4 Durability.- The camera shall be capable of providing the cycles of operation and functioning under the conditions specified herein, when tested in accordance with Section 4.

3.25.4.1 Camera.- The camera shall be capable of operating for 20,000 cycles, continuously, with a maximum of thirty (30) minutes permitted for each changing of film in the film cassettes; in any selected mode or modes of operation using the IMC rates and exposure rates specified in this specification. Cleaning and minor adjustments which will require no more than ten minutes for both, will be permitted, if required, after every 250 cycles. The replacing or removing of parts are not permitted. The ten minutes, permitted for cleaning and minor adjusting, will be part of the thirty minutes allowed for film changing. At no time, during testing, will the camera be idle for more than thirty minutes.

3.25.4.1.1 Camera body.- Other than that specified in 3.25.4.1, the camera body shall not require maintenance for 20,000 cycles.

3.25.4.1.2 Lens cone.- The lens cone, including shutter, shall not require maintenance for 20,000 cycles. The 20,000 cycles shall be continuous, however, thirty minutes of nonoperation will be permitted during the changing of film as specified in 3.25.4.1.

3.25.5 Film damage.- The film shall not be damaged by its passage through the camera. The film shall not suffer tears, distortion, abrasions, scratches or other deleterious markings that would affect the quality of the photographic image.

3.25.6 Light tightness.- The cassette and camera design shall be such that no more than 2 feet of film will be fogged upon removal of a loaded cassette from the camera body, with the cassette exposed to bright sunlight for one hour.

3.25.7 Camera resolution.- The complete camera, when tested in accordance with Method 11 of MIL-STD-150, with the shutter immobilized in an open position and the lens at maximum aperture, shall produce an AWAR of not less than 89 percent of the AWAR of the individual lens. (See 4.9.)

3.26 Camera control system compatibility.- The camera shall provide the functions specified herein when supplied with drive signals, control functions, and power inputs as obtained from the Mohawk Photo System Unit.

3.26.1 Control system signals.- Drive signals for movement of film and adjustment of exposure factors shall be as specified herein.

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3.26.1.1 IMC and film drive.- Power to the drive motor of the camera shall be supplied from a suitable velocity servo amplifier loop. The amplifier shall be capable of delivering a minimum of 2.5 amps at 85 volts d.c. for maximum motor speed, and a minimum of 5.0 amps during stall condition. The loop shall be calibrated to accomodate an analog IMC feedback voltage from the camera of 7 volts d.c. per inch per second of IMC.

3.26.1.2 Exposure control.- The camera shall be provided with an exposure reference source of nominally 115 volts rms 400 cps ac. Power to the exposure drive motor shall be supplied from a suitable positional servo amplifier loop. The exposure drive amplifier shall be capable of providing 400 cps ac voltages to the control winding of the drive motor nominally 90 degrees out of phase with the exposure reference voltage. Phase relationship of the control voltage shall be leading the reference voltage to correct for the condition of increasing light level, and lagging the reference of decreasing light level. The amplifier shall be capable of providing sufficient output to drive the loop to within 0.7 v rms maximum error voltage from a feedback source impedance of 4000 ohms. Command voltages to the loop shall be in accordance with the following relationship which expresses the analog brightness feedback voltage from the camera:

$$E_b = \frac{E_s}{10} \log_2 B \text{ v rms 400 cps ac}$$

where E_b = camera feedback voltage

E_s = exposure reference source voltage

B = brightness in foot-lamberts to which camera exposure parameters are adjusted.

3.26.2 Control functions.- The camera shall be provided with electrical control impulses from circuitry capable of supplying 1.5 amp. dc inductive minimum load as specified in the following paragraphs.

3.26.2.1 Operate command.- Operate command shall be provided by closure of a single circuit to ground.

3.26.2.2 Cycle pulses.- Plus 28 v dc pulses of 75 milliseconds (plus or minus 25) duration shall be provided to initiate shutter trip during pulse and pulse IMC modes of operation.

3.26.2.3 Shutter close pulses.- Plus 28 v dc pulses of 75 milliseconds (plus or minus 25) duration shall be provided for "capping shutter" initiation during night mode of operation.

3.26.3 Power inputs.- DC and 400 cps ac power shall be provided to the camera as specified herein. (See 3.22.1).

3.27 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

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3.28 Lubricants.- Lubricants shall be selected from the Lubricants Guide, Drawing SC-A-46469. Two or more lubricants shall not be used where one will meet the requirements of the item in which they are used. If none of the lubricants listed by the Guide will meet the requirements of the item in which they are used, the contractor shall propose an alternative lubricant for approval and shall furnish the information listed in the General Note of the Guide; in addition, a test sample of the proposed lubricant shall be furnished when required by the Government. Contact shall be avoided between lubricants and materials (such as certain paints, rubbers, and plastics) on which the lubricants have a solvent effect.

3.29 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.30 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein;

3.11	Cleaning	3.19.3	Protection for wires and cables
3.14	Riveting	3.19.5	Clearance in wiring and cabling
3.15	General requirements for securing of parts	3.19.6	Splicing of wires
3.16.2.2	Flux and cleaning agents for soldering	3.19.8	Connections, general
3.16.2.3	Process for soldering	3.19.9	Grounding, general
3.18.1	Cleaning of welding		
3.18.2	Process for welding		
3.19	Wiring and cabling	3.10.3	Self-tapping screws
3.19.2	Slack in wires and cables		

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5.)

(2) Acceptance inspection of preparation for delivery. (See 4.26.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables IV, V and VI respectively). The environmental tests will normally be performed in the order specified in Standard MIL-STD-152; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table III. - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req. Para.	Insp. Para.
Acceleration		
Equipment Operating	3.23.1e	4.6.1
Equipment Nonoperating	3.23.2g	4.6.2
Attitude	3.23.3	4.7
Drip-proof	3.23.4	4.10
Explosion proof	3.23.1g	4.12
Fungus	3.23.2d	4.15
Moisture resistance	3.23.1c	
	3.23.2c	4.18
Radio interference	3.23.1f	4.19
Shock	3.23.2	4.20
Temperature-Altitude	3.23.1a, 3.23.1b	
	3.23.2a, 3.23.2b	4.23
Vibration	3.23.1d	4.24

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4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Diaphragm	3.6.1.3
Lens, 6-inch focal length	3.6.7
Filters	3.7
Finish	3.12
Marking	3.13

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government-- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table IV and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.8) shall be last. (See 6.4.)

Table IV. - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.30	4.25		
Camera Body LA-134( )	3.5	4.25	1%	4%
Cassette, film (100 ft capacity LA-135( )	3.8	4.25	1%	4%
Lens Cone, 6-inch focal length, LA-131( )	3.6	4.25	1%	4%
Operational				
Camera, Still Picture KA-30( )	3.24	4.8	1%	*
Exposure Feed-back	3.6.5.1	4.13	1%	*
Film Advance	3.5.1 thru 3.5.3 3.25.3	4.14	1%	*

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Table IV.- Group A Inspection (Continued)

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Operational (Cont'd)				
IMC Drive Rate	3.24.1 3.24.3 3.24.4 3.25.2	4.16	1%	*
Shutter speed	3.6.1.1	4.21	1%	*

* All defects are considered major.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table V and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in table V, shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
4.0%	L-8	L-6
4.5%	L-7	L-5

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table V.- Group B inspection

Inspection	Req. Para.	Insp. Para.	A Q L (See 4.5.2.1)
Camera resolution	3.6.7.1 3.25.7	4.9	6.5%
Interchangeability	3.27	4.17	4.0%

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4.5.3 Group C inspection.- This inspection shall be as listed in table VI, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table VI.- Group C inspection

Inspection	Req. Para.	Insp. Para.
Durability	3.25.4	4.11 *
Solderless terminal-lug Connections	3.19.8.2	4.22

* Sample units subjected to this inspection shall not be furnished on contract.

4.5.3.1 Sampling for inspection of equipment.- One Camera, Still Picture, KA-30( ), for each group C inspection shall be selected in accordance with table VII below, without regard to its quality.

Table VII.- Sampling for Group C inspection

Quantity of Equipment on Contract	Quantity Requiring Inspection
0 to 50 inclusive	1 unit from first 25 produced
0 to 100 inclusive	1 unit from first 50 produced
0 to 500 inclusive	1 unit from first 250 produced, and 1 unit from second 250 produced
0 to 1000 inclusive	1 unit quarterly (i.e. every three months)

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.3.)

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4.5.4 Reinspection of conforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Acceleration.-

4.6.1 Equipment operating.- The equipment shall be tested in accordance with paragraph 4.4.7 of Standard MIL-STD-152 to determine compliance with paragraph 3.23.1e.

4.6.2 Equipment nonoperating.- The equipment shall be tested in accordance with paragraph 4.4.8 of Standard MIL-STD-152 to determine compliance with paragraph 3.23.2g.

4.7 Attitude.- Camera, Still Picture, KA-30( ) shall be operated in at least six different attitudes and positions from horizontal to vertical, to determine compliance with the requirements specified in 3.23.3.

4.8 Camera operation.- Using a cassette loaded with film as specified in 3.8.2, Camera, Still Picture KA-30( ) shall be operated in all modes as specified in paragraph 3.24. The resulting photographs shall conform to the resolution requirements specified in paragraph 3.25.7. The number of cycles incurred by this test shall be kept to a minimum and will be the number of cycles showing on component cycle counters as delivered.

4.9 Camera resolution.- Camera, Still Picture, KA-30( ), assembled for operation, shall be tested in accordance with Method 11 of Standard MIL-STD-150. The lens shall be at maximum relative aperture with the shutter immobilized in an open position and the film held in the magazine by vacuum during the test. The exposure shall be made by controlling the duration of the target illumination. The results of this test shall show compliance with the requirements specified in paragraph 3.25.7.

4.10 Dripproof test.- Mount the camera to simulate service installation and perform the following test:

Using a domestic type sprinkling faucet on the outlet of a nozzle having a pressure head of one foot of water and located three feet from the camera, allow water to fall upon the camera at a rate of five gallons per square foot of top surface area during a period of five minutes. At the end of the five minute period, the cassettes and lens cone shall be disassembled from the camera body and the parts inspected for any evidence of water internally. (See 3.23.4.)

4.11 Durability.- Durability tests shall be as specified herein without failure or loss of capability to meet the requirements of this specification.

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4.11.1 Camera body and cassete.- The camera body shall be operated for a total of 20,000 cycles, as specified in 3.25.4.1, while transporting 100 foot and 250 foot lengths of film from a initially full cassette tc an initially empty take-up cassette. During the test, the camera body shall be operated in the various modes, 5000 cycles in each mode of operation, as specified in paragraph 3.24. No maintenance other than cleaning and minor adjustments as specified in 3.25.4.1 shall be permitted during the test.

4.11.2 Lens cone (see 3.25.4.2).- The shutter and data recorder shall be subjected to 20,000 continuous cycles of operation without maintenance or adjustment. The shutter speed and iris diaphragm opening shall be varied during the test.

4.12 Explosion proof test.- The explosion proof test shall be performed in accordance with Explosion Proof Test, Procedure I of MIL-E-5272 except that a sampling chamber shall be used to determine if any explosive mixture is present. No external explosion shall occur while the equipment is being operated in an explosive atmosphere. Internal explosions shall be permitted only if they are contained within flame arresting enclosures of the equipment and if they do not impair proper operation of the equipment. The sampling chamber shall be complete with blower, valves, and spark plug. The temperature within the chamber during the test procedure should not exceed plus 20 degrees to 50 degrees C(68 degrees to 122 degrees F). (See 3.23.1g).

4.13 Exposure feed-back.- The lens cone shall be tested to determine compliance with the requirement specified in 3.6.5.1.

4.14 Film advance test.- Each camera body and cassette to be supplied shall undergo this test. The camera shall transport 100 feet and 250 feet of film while operating in all specified modes throughout the specified operating limits. A light source shall be used to expose or fog film in the format area during this test. The film shall be developed and inspected for scratches, damage, frame spacing and recording of data.

4.15 Fungus.- The equipment shall be tested in accordance with paragraph 4.4.13 of Standard MIL-STD-152 to determine compliance with paragraph 3.23.2d.

4.16 IMC drive rate.- Camera body LA-134( ) shall be tested to determine compliance with paragraph 3.25.2. During the test, the camera body, operating in the autocycle mode(see 3.24.1), shall transport 100 foot and 250 foot lengths of film loaded in cassettes and shall be supplied with a vacuum source. The IMC rate accuracy shall be determined by measuring total cycle times resulting when the camera is driven to produce tachometer output voltages corresponding to the IMC rate capabilities of the camera.

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4.17 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.27. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.18 Moisture resistance.- The equipment shall be tested in accordance with paragraph 4.4.11 of Standard MIL-STD-152 except that total number of cycles shall be 10 instead of 15 cycles.

4.19 Radio interference test.- The complete camera shall be tested for compliance with MIL-STD-152 and MIL-I-6181 except that once per cycle, radio interference conducted or radiated, may exceed 20 decibels above the limits of MIL-I-6181 in the frequency range of 0.15 to 150 megacycles for 0.2 seconds. Tests for conducted radio interference shall be performed on the 28 v dc and 115 v 400 cycle power lines. Susceptibility tests are not required. A Ferris Model 32A Radio Noise Test Set shall be used for all measurements of radiated and conducted interference between the limits of 0.15 to 20 mc and an Empire Device Model NF-105 (AN-URM-7), Radio Noise Meter shall be used for measurements of radiated interference between the limits of 20 mc to 150 mc.

4.20 Shock.- The equipment shall be tested in accordance with paragraph 4.4.10 of Standard MIL-STD-152 to determine compliance with the requirement specified in paragraph 3.23.2f.

4.21 Shutter speed.- The shutter speed of the lens cone shall be tested to determine compliance with the requirement specified in paragraph 3.6.1.1.

4.22 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.19.8.2.

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4.23 Temperature.- The equipment shall be subjected to the following tests as specified in Standard MIL-STD-152.

MIL-STD-152	Test
4.2.2	High temperature (equipment operating)
4.2.3	High temperature (equipment nonoperating)
4.2.4	Low temperature altitude (equipment operating)
4.2.5	Low temperature altitude (equipment nonoperating).

4.24 Vibration.- The same camera using both the 100 foot cassette and the 250 foot cassette, shall be tested in accordance with paragraph 4.4.9.2 of Standard MIL-STD-152 and Procedure I of Specification MIL-E-5272, except that tests shall be conducted at room temperature and the applied acceleration shall be as follows:

a. Using the 100 foot cassette, the applied acceleration shall be ± 5G in lieu of ± 10G.

b. Using the 250 foot cassette, the applied acceleration shall be:

(1) ± 2.5G from 40 cycles per second to 270 cycles per second in lieu of ± 10 G.

(2) ± 5G from 271 cycles per second to 500 cycles per second in lieu of ± 10G.

During the test, the camera shall be operated in the autocycle mode at a rate of 1 cycle in 5 seconds except that at the beginning of the resonant periods, the camera shall be operated over the full range of cycle rates and in all modes of operation.

4.25 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standards MIL-STD-180 and MIL-STD-252.

4.26 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with the applicable provisions of Drawing SC-A-362100.

6. NOTES

6.1 Intended use.- Camera, Still Picture, KA-30( ) is used in aerial reconnaissance aircraft for producing photographic negatives of terrain, or objects thereon.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(d) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.29.)

(h) Quantity of tools and running spare parts required. (See 3.29.)

(i) Submission of the statement of treatment referenced in 3.17, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3.)

6.4 Location of operational inspection.- It is desirable that the operational inspection (4.8) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: KA-30W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

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6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below;

(a) Type A---The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B-- That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--- A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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